

SECURING THE FUTURE

Scaling Engineering Innovations for Critical Infrastructure Resiliency and National Security in Nigeria

Guest Lecture

2026 NSE Ibadan Annual Conference and Exhibition

Ifedayo Akintunde Engineering House, Kolapo Ishola GRA, Akobo, Ibadan

Wednesday, 19 August 2026

Dr. Olusola Sayeed Ayoola, MIET, MNSE, MNCS

Founder and Chief Executive Officer

Robotics and Artificial Intelligence Nigeria (RAIN)

Protocols

The Chairman of this occasion, the Chairman of NSE Ibadan Branch, Engr. Ayokunnu Olukunmi Ojedeke, MNSE, the Chairman of the Conference Planning

Sub-Committee, Engr. Prof. Wasiu Olabamiji Ajagbe, FNSE, FNIEEE, members of the Branch Executive Committee, distinguished Fellows, my fellow engineers, invited guests, gentlemen of the press, ladies and gentlemen.

All protocols duly observed.

It gives me great pleasure to speak at the 2026 Annual Conference of the Nigerian Society of Engineers, Ibadan Branch.

We must begin by reminding ourselves what we are. We are engineers. We are problem solvers. Where there is a problem, there is a need for a solution. And the best solutions are generated internally. They are adapted to the environment. They last, provided the engineers who build them look inward and design for the environment they are actually in.

That single sentence carries the whole weight of today's theme. Because it also means the opposite is true. If engineers continue to import solutions, and all we do is swap and swap with what has already been done abroad, then we will continue to find ourselves in trouble day after day.

So in the bid to internalise our problems and to localise our solutions, we are gathered here once again. And it is fitting that we are gathered in Ibadan. This branch is the Premier Branch. We have historically been the forerunner when it comes to building solutions that last, whether the work is electrical, industrial, environmental, civil or agricultural. As the world moves towards smart solutions, the Internet of Things, robotics and artificial intelligence, Ibadan Branch is not left out. I represent this branch proudly wherever I go as an engineer. I am honoured to be given this podium.

The theme of this conference is apt and timely: Securing the Future: Scaling Engineering Innovations for Critical Infrastructure Resiliency and National Security in Nigeria.

Nothing can be truer than this. If we do not scale the innovations we build locally for the critical infrastructure we need, we will not find buyers. We will not find a market even here, in the very place where that infrastructure is critical. And if we cannot find buyers for critical infrastructure at home, then we are exposing our national need to foreign solutions, with their own embedded threats.

Foreign solutions do not only come with threats. They come with a cost we cannot afford. For a nation that is a consumer nation, we will simply continue to import

solutions while selling the God-given natural resources that were supposed to be our leverage for the future. And at the end of that road, the future is not secured. It is mortgaged.

Let me put numbers to the urgency, so that we are not speaking in abstractions.

Since 2010, Nigeria has recorded hundreds of partial and total national grid collapses. Twelve of those happened in 2024 alone. Our installed generation capacity is quoted at somewhere between 13,000 and 15,500 megawatts. What we actually deliver to homes and factories rarely exceeds 5,000. Experts put the minimum requirement for adequate national supply at around 30,000 megawatts. We cannot reliably transmit one-sixth of what we need.

And it is not only power. In the first six months of 2026 alone, the Nigerian Communications Commission recorded over five thousand fibre optic cable cuts from road construction, excavation and related civil works, in a country where telecommunications infrastructure has already been formally designated as Critical National Information Infrastructure since 2024, and where damaging it already carries a ten-year jail term.

So we already have the law. What we do not have is the engineering.

Every blackout is a security incident. Every fibre cut is an economic attack. Every collapsed bridge is a denial-of-service on the movement of goods, people and troops. This is why infrastructure resilience is not a utilities conversation. It is a national security conversation. And it is our conversation, as engineers.

So how do we secure this future? The theme has told us what to do. What remains is for us to break it down honestly. What exactly is the situation today?

3. The Diagnosis: We Are Working in Silos

Let me say to you plainly that the situation today is that we are all working in silos. Engineers. Lawyers. Politicians. Bankers. Everybody is in a silo.

It is partly the nature of mankind. We are political. We are competitive. But what serious organisations abroad have realised is that the best only comes out when we collaborate.

And collaboration is not only between people in the private sector. Collaboration is also among government agencies. It is among the tiers of government. And it is between government and the private sector. So ask yourself honestly: how well are we cooperating?

That is the first step we must imbibe. Cooperation. Where the banks support the businesses, believing in them, trusting their ideas, supporting them in the areas where engineering companies are not strong, marketing for instance, and foreign partnerships. It is not the job of the engineer to go looking for partners. That is the job of the business investor. So how well are our investors trusting our local talent?

I am also speaking to government. Agencies of government competing with one another, refusing to share their data, refusing to even share their problems. That is bad. There is an engineer with an innovation, and the government officer who is supposed to use it cannot adopt it, because adopting it would look like an admission of weakness, or because sharing the problem feels like exposure. That is not how it should be. Corporates run open systems. Many people are doing the same thing repeatedly and do not know that somebody else has already done it.

Consider the matter of research publication. Most of our very brilliant researchers publish in foreign journals. For that reason you cannot access their work locally. For that reason we cannot build on what they have done. Your citizens cannot read it because they must pay. But foreigners are sponsored to access it, so they build on your knowledge, while at home we rebuild the same thing over and over again.

Our best research sits behind a paywall that our own citizens cannot cross.

The point I am making is that platforms must be built to encourage collaboration. And the Nigerian Society of Engineers has the mandate to look at this. Let it start from the Society itself. How do we interface with the Nigerian Medical Association? How do we interface with the Manufacturers Association of Nigeria? Some of this is already happening because we are interwoven. But we must be deliberate about it.

4. An Honest SWOT

Let us be honest with ourselves in the way engineers are trained to be honest. Let us look at strengths, weaknesses, opportunities and threats.

4.1 Strengths

Nigeria has strength. We are gifted in our youthful population. Everywhere you go, people are ready to work. We have resources, if we are willing to drill and to refine. We have a comparatively affordable labour force, if we are serious about deploying it. We have fewer regulatory barriers to entry than many countries. We have a favourable topography and climate, and a natural safety from many of the catastrophic events that other nations must engineer against. And above all we have brilliance. Nigerian engineers are found at the top of institutions across the world.

I say the following as encouragement, not as flattery, because we have a domestic proof that local capacity is real and can be built by policy. Before the year 2010, Nigerian participation in our own oil and gas industry was below five per cent. Today, following the Nigerian Oil and Gas Industry Content Development Act, that figure stands at over sixty-one per cent, with more than eight billion dollars a year now retained in-country rather than exported. Earlier this year, TotalEnergies announced ninety-five per cent Nigerian content on one of its projects, and said plainly that it proves local expertise can deliver complex work to international standards.

Under five per cent to sixty-one per cent in fifteen years. Not by prayer. By law, by enforcement, and by deliberate capacity building. If it worked for oil and gas, there is no engineering reason it cannot work for power, transport, water, telecoms and defence.

4.2 Weaknesses

But there are weaknesses, and I will not soften them.

The first is corruption at scale. Corruption is not our problem alone. Different countries have their own ways of dealing with it. But here it has become a way of life. I wrote in a newspaper in 2005, twenty-one years ago, that corruption had

become a way of life in Nigeria, so that if you are not corrupt, you are the one who looks odd.

And it distorts engineering itself. Consider a simple case. You build a breathalyser to help road safety detect drunk drivers, automatically flag them and process the penalty through the system. It works. It is cheap. It saves lives. And it will not be adopted. Why? Because some would prefer to retain the personal power to forgive on the road. Bring an innovative idea and you will hear expansive excuses for why it should not work, or why it should not be allowed to work, in the private sector and in the public sector. You all know this.

The second weakness is weak purchasing power across a large part of the population. This is not the fault of any single administration. It is perennial. Governments keep trying, and there will still be a proportion of the population that cannot afford. So your idea is great, but it is expensive. Where does that leave us?

The third weakness, and this one is ours to own as a profession, is that we have not built the culture of maintenance. We build and we do not maintain. Roads, plants, rail, transformers. Some of the transformers on our network are older than most of the engineers in this room. There is a rule of thumb in infrastructure economics that every one naira of maintenance deferred becomes roughly four naira of reconstruction later. We defer, and then we are surprised by the bill.

4.3 Opportunities

Now, in all of these, there are opportunities. This is where I want you to lean forward.

People say power supply is bad, so they cannot do business, cannot manufacture, cannot produce, and the cost of living and of doing business is high. All of that is true. And every line of it is a specification for a product.

- Solar and storage. Not just panels. Infrastructure and appliances designed to run on intermittent supply.
- Energy management, load prediction and demand forecasting for estates, campuses and industrial clusters.
- Fault-line detection, surge protection and automated fault location on distribution networks.

- Predictive maintenance for manufacturing plants, so downtime is scheduled rather than suffered.
- Road condition data collection. Imagine if every commercial vehicle in this city carried nothing more than a gyroscope and an accelerometer. As they drive they detect potholes. When they reach the bus stop and pick up internet, the data synchronises. Now government has a live map of which roads are bad and which of those carry the heaviest traffic. Google already has that data. Why should we not have it ourselves?
- Trade and logistics platforms that enable intra-national and intra-African trade under our own laws.

Every one of those is a business. Every one of those is a national security asset. Because a country that cannot see its own infrastructure cannot defend it.

4.4 Threats

And then there are threats to those opportunities.

It is cheaper to buy foreign than to buy local. Somebody abroad will build it cheaper. Somebody abroad has done it before, and the contract will go to them. The expatriate is presumed to do it better, and so you will not get the opportunity to prove otherwise.

The second threat is that our practical skill level is low. Not our theory. Our practice. Let me tell you a true story. A very accomplished Nigerian lady returned home from the United States. She came back with an industrial robot arm, the kind used in manufacturing. She could not find anybody to operate it. She came to us at RAIN. And even we hesitated. Why? Because that arm costs in the region of three hundred million naira, there is no insurance cover for it, and if we damage it in the process of learning, who pays? So the honest answer was that we were not ready to take that risk. If we had one in our own laboratory, we would have been practising on it for years. But we do not have one. She has seen the opportunity and she is tapping into it. But the threat is exposed: we do not have sufficient hands-on skill, because we do not have sufficient hands-on equipment.

The third threat is brain drain. You cannot pay your workers enough. They see opportunities abroad and they take them, and we are left worse off than before.

The fourth threat is data. We continue to use foreign applications that have access to our data, while starving local applications of the very data they need to become

good. Data is the raw material of the twenty-first century, and we are exporting it free of charge, the same way we exported everything else.

5. Who Is Doing What, and At What Scale?

Now to be fair, some people are trying, but at what scale.

5.1 The Researchers

First, our teachers in the universities with their years and years of research and international Conferences abroad. They have seen what others are doing and they are trying within their own space, mentoring students. They are trying. But is it enough? At what scale are they creating impact?

They create impact in two ways: they build the talent that will do the work, and they advise the businesses that will adopt it. But the talent graduates and the business does not employ them. There is a disconnect. The business asks, did they teach you how to do it? And the student says, no, but they told us how to do it.

***They told you how to do it. They did not teach you how to do it.
Which means they did not do it with you.***

We do not have the equipment, and so we are back to square one. The company brings in an expatriate. The student goes online to look for a course that will teach him how to actually do it. Our professors are trying, but we must equip their laboratories so that they can build what they know how to build, and demonstrate it for the young people coming behind them.

And where a university cannot be equipped, let us encourage facilities like RAIN that are already doing it, so that such facilities exist across the country. I give kudos to the universities that are investing in infrastructure. But please, let us open these facilities to the students. I know it is expensive. If they destroy it, who refunds it? That is a real question, and it deserves a real scheme: how do we fund our research facilities, how do we fund our researchers, and how do we insure our laboratories?

This is where the Triple Helix comes in: government, academia and the private sector. It is a tested model for moving innovation from the gown to the town, and for pulling the next generation into the pipeline behind it. This is how DARPA

operates in the United States. It is how Fraunhofer operates in Germany, where dozens of applied research institutes run on a deliberate funding formula, roughly one-third base public funding and two-thirds earned from contract research, which forces them to stay relevant to industry. It is how ITRI operated in Taiwan, an applied research institute that did not merely publish papers but spun out TSMC and UMC, the companies that now make the chips the whole world depends on.

Taiwan did not become a semiconductor superpower because Taiwanese people are smarter than Nigerians. It became one because a government-funded applied research institute was mandated to spin out companies, not to publish papers.

5.2 The Youth

The second group doing their best with the little they have are the students and the young people. Those in the universities, those in the polytechnics, and those who are not in school at all.

I must commend them. How many times have we seen university students execute final year projects funded from their own pockets, with their parents supporting them because the family believed in them? How many times have we seen young people who are not even in university build things that are worthy of news headlines? Nigerian boy builds a drone. Nigerian students build an electric bus. Nigerian student creates an app that speaks multiple local languages.

But once they showcase the talent, that is where it ends. After the headline, what happens? At most, a handshake with a governor or a president. Perhaps a scholarship abroad. And once you send them abroad, you may never see them again.

I remember the young man who built a drone and was taken to Finland by a Finnish company. On the day I heard that story, I understood something. I am not saying exporting talent is bad in itself. But what is our process for rebuilding more talent behind the one that left? Because the future we are trying to secure is in these young people.

And let me speak to the young people directly, because time is the one resource engineering cannot manufacture.

You enter the engine room of national development at about twenty-six or twenty-seven. Take a working life of about thirty-five years. That means you retire around sixty-something. So you have roughly three decades to make an impact.

- The first decade you spend learning how to do it, and how not to do it.
- The second decade is where you actually make impact. You chart the path. You solve the problems.
- The third decade is where you make sure there is something to hand over: a team that can continue, and a solution that can outlive you.

Thirty years. That is all. So to the young man or young lady about to graduate: time is of the essence. Whatever you are going to do, there is no time. Please do not waste it.

5.3 The Business Owners

The third group are the business owners and the service providers. They see the problem day to day. They understand the challenges. They know what solutions are required. But they are not equipped to build those solutions. They are only equipped to buy them when they are ready.

So they go looking for talent, looking for solution providers, and most times the safest bet appears to come from abroad, because they are looking for people who have done it and tested it. They cannot afford to take a risk. They are in business. It must work the first time.

These people need to rethink their strategy. And I say this as one of them. We must not continue to eat only already-prepared meals, unless we know that they were prepared locally.

You want to build a refinery, and somebody comes from Russia or Germany or China and says, I can build it for you. Fine. But say to them: I want it built by my own people. Take them. Train them. I like what Dangote did with the refinery. But the honest question remains: how much real training is transferred? This is precisely the job that NOTAP, the National Office for Technology Acquisition and Promotion, is meant to enforce, ensuring there is proper handover from the expatriate to the local engineer. Otherwise we end up with a situation where they

are not transferring knowledge at all. They are only teaching us how to service, so that we must always call them back.

And I understand the fear of the Nigerian business owner. We worry about the Nigerian factor. We worry that our own people will underperform, will not deliver to standard even when they are capable. But is it not the same Nigerian who misbehaves at home and behaves impeccably the moment he lands abroad? Then the problem was never that they do not know how to behave. The problem is the system they are behaving in. So our question is not whether Nigerians can perform. It is what system we must build so that performance is the only rational option.

Let us also be clear-eyed about our celebrated successes. Which local solution are you consuming today? In fintech we know Paystack. We know Flutterwave. Nigerian ideas, brilliant ideas. But largely funded from abroad, and in significant part registered abroad, operating under laws that are not Nigerian laws. When a Nigerian company raises millions or billions of dollars, those funds come from foreign investors who have ensured the business operates under foreign legal protection. We then have very little say in it. For many founders the moment of joy is the moment they raise from abroad. Some of us are not built that way. We believe in local capital, local law and local ownership, and we intend to remain patriotic and committed to local development through local sources and local opportunities.

6. The Missing Link: Nobody Will Let You Test

So where are we in this whole scheme of things? Can we compete with foreigners today, as we are? No. Not yet. And I want to name precisely why, because the reason is not talent and it is not intelligence.

The reason is that in Nigeria, there is nowhere to test.

Let us believe we can do it. Let us not assume the knowledge is not here. Let us not wait for validation from foreigners. If you see a local talent, adopt the person. If you see a local researcher, adopt their ideas. If you see a local business, support and invest in what they are doing, and equip the researcher to do more. And let us

not be afraid to make mistakes, because if we never make mistakes, where will the knowledge and experience to do better come from?

Go to the Nigerian military and tell them you have a solution that can counter insecurity. They will tell you to go and test it first. But how do you test a military-grade solution when you are not a terrorist? Do you have a licence to test? Even if the military gives you cover, there are hurdles before you get to the point of testing. Are you certified to military standard? Getting to that point requires collaboration, and it requires an institution.

It is the same with the banks. Bring a solution that detects cyber-security threats and they will tell you to prove it at a lower risk level first. But a system that works at low risk can behave completely differently under the pressure of the real environment. So you are stuck between two impossible instructions: prove it in the real environment, and stay out of the real environment.

I remember a man who claimed he had built an AI system better than Google. They said, fine, test it in Nasarawa. And when he tried, he said, sorry, there is no internet, I am actually using Google. You are using Google and you are telling us you are better than Google.

There is a name for this gap in the innovation literature, and I want the young engineers here to write it down.

6.1 The Valley of Death

Engineers and funding agencies worldwide measure how mature a technology is on a scale of one to nine, called Technology Readiness Levels. TRL 1 to 3 is basic research and proof of concept. That is the university. TRL 8 and 9 is a proven, qualified system operating in the real world. That is the market. Between them, TRL 4 to 7, is where you validate in a relevant environment, demonstrate in an operational environment, and prove the system under real conditions.

Nigeria funds TRL 1 to 3. Nigeria buys TRL 8 to 9. Nigeria has almost nothing in between. The whole of TRL 4 to 7 is missing. That is why our research dies in the laboratory and our procurement lands in a foreign account.

That gap is what the literature calls the valley of death. And every country that industrialised built an institution to carry its innovations across it. Ours is not missing because it is expensive. It is missing because we have never named it as a thing we are supposed to build.

6.2 The Answer: Build the Sandbox

So my central proposal to this conference is this. We must build what I call the sandbox.

A sandbox is a faithful similitude of the real operating environment, in which anything can be attempted under real conditions, but without real consequences in the real world.

Let me make it concrete. Suppose you build a system that automatically marks handwritten WAEC scripts. You tell WAEC, use my system. WAEC cannot possibly rely on it for a live examination, and they are right not to. But WAEC can build a platform where the same traffic that comes to them is mirrored into your system, and we see how you handle it: how fast, how accurately, at what failure rate. You mark those scripts. Those marks are never issued to any student. They exist purely as a test.

That is not a Nigerian improvisation. That is a recognised engineering practice with a formal name: shadow deployment, or parallel running. The candidate system runs alongside the live system on live data, and its outputs are compared but never acted upon. It is how banks migrate core systems. It is how air traffic control validates new software. It is how we should be qualifying every locally built system in this country.

Now apply that thinking everywhere:

- Defence and security: a controlled range and a simulated threat environment where a Nigerian counter-drone or surveillance system can be qualified without a live terrorist.
- Banking and payments: mirrored transaction streams for fraud and intrusion detection, at production volume.
- Power: a hardware-in-the-loop testbed where a real relay or controller is wired into a simulated grid, so it can fail a thousand times before it ever touches a live feeder.

- Manufacturing: a shared demonstration line where predictive maintenance and robotics can be proven on real machinery.
- Education and health: shadow-mode evaluation of assessment, diagnostic and triage systems on real data with no real decisions.
- Transport and roads: an instrumented corridor in this state where road-condition sensing and traffic systems are validated at scale.

6.3 And Here Is the Good News

The sandbox is not an untested idea that I am inventing on this podium. It is an established regulatory instrument, and Nigeria already operates one. We simply built it in finance and never extended it to engineering.

The United Kingdom's Financial Conduct Authority created the first regulatory sandbox in 2016. More than sixty countries have since adopted the model, and by one World Bank assessment Africa now hosts the third-highest number of sandboxes in the world. The Central Bank of Nigeria issued its own Framework for Regulatory Sandbox Operations in January 2021. The Securities and Exchange Commission runs an Accelerated Regulatory Incubation Programme.

As I speak to you this morning, applications for Cohort 2 of the Central Bank of Nigeria's Regulatory Sandbox are open. They opened on 12 August and they close on 31 August 2026. A Nigerian fintech can legally test an unlicensed product on live money under supervision. A Nigerian engineer cannot legally test a sensor on a live transformer. Explain that to me.

If the Central Bank can do it for money, the Nigerian Electricity Regulatory Commission can do it for the grid. The Nigerian Communications Commission can do it for networks. The Office of the National Security Adviser can do it for critical national infrastructure. The Standards Organisation of Nigeria and COREN can define the qualification pathway. And the Nigerian Society of Engineers can be the body that writes the proposal and puts it on the table.

6.4 The Tools of the Sandbox

In creating this sandbox, several technologies become directly useful, and this is where our young engineers should be building careers.

Simulation environments, which let you model a system before it exists. Digital twins, which are live virtual replicas of a real physical asset, continuously fed by sensor data, so that you can test a change on the twin before you touch the asset. Singapore built a digital twin of the entire city state. The United Kingdom runs a National Digital Twin Programme with a published set of principles for how such twins should be governed. Virtual and augmented reality, for training operators safely on equipment worth hundreds of millions of naira. Hardware-in-the-loop rigs. And yes, game development, which is not a triviality; the same physics and rendering engines now drive robotics simulation, surgical training and even aspects of drug discovery.

These are buildable here. Some of them we are already building here.

7. Adapt, Do Not Merely Import

Let me now turn to a discipline that I believe is the true signature of a mature engineering profession: adaptation.

Take the Internet of Things. In principle, every transformer, every feeder, every line on the grid should be able to notify us when there is a failure, when there is an outage, when supply is restored and at what time it left. That gives us fault analysis, fault location, load management, demand prediction, and beyond that, population and planning data.

Now, when I say this, someone will correctly point out that this already exists. It has been done in laboratories. It has been done in schools. But for a business or a utility to adopt it, they look at the real environment, and the real environment says: rural areas have little or no internet; equipment gets vandalised; communities may resist and destroy what they do not understand; and there is no reliable power to keep the monitoring system itself running.

Every one of those objections is an engineering requirement, not an excuse.

- No internet at the site? Fall back to GSM. No GSM either? Then store and forward. Let the node buffer its data locally, and synchronise whenever any passing device or maintenance vehicle raises a hotspot. Is that possible? It is entirely possible.

- Vandalism? Then design for tamper detection, low resale value of components, potted enclosures and mesh redundancy so that losing one node does not blind the network.
- No power? Then design for microwatt duty cycling and solar harvesting, and accept a lower sampling rate as a deliberate trade.
- Community resistance? Then engineer the community in. Give them a reason to protect the node, because the node also tells them when their power is coming back.

Solutions like this have not been built before, not because they are not thinkable, but because they look like unnecessary stress in an environment where they are not needed. In a foreign environment, they are not needed. Here, they are the whole job. That is the entire argument for local engineering, in one paragraph.

Let me give you an example of adaptation from somewhere else, so you can see how ordinary it really is. In the United Kingdom, when you want to cross the road, there is a traffic light for cars and a light for pedestrians. In busy areas, pedestrians simply wait for their green. But in quieter towns, you do not want to be stopping traffic for pedestrians who may not even be there. So what did they do? They put a button on the pole. Only when a pedestrian presses it does the pedestrian phase run at all. Otherwise the crossing stays red throughout.

That button is not new technology. It is an adaptation to reality. That is all engineering is.

And this is the kind of adaptation we must apply to every technology that arrives here. Language: we must adapt systems to Yoruba, Hausa, Igbo, Pidgin. Weather: much of the electronics and many of the vehicles sold here were never designed for our humidity, our dust, our harmattan or our temperature cycles, and our engineers spend their lives repairing failures that were designed in elsewhere. The answer is not to complain about the equipment. The answer is to specify, and eventually to build, for our own conditions.

8. And Then, Market It

I want to close on something engineers are usually very bad at.

If you create a sandbox and you prove your solution, the business case will come. I am confident of that. But you must also believe in marketing. As engineers, let us market. Let us take advantage of social media, of advertising, of creative thinking to push our solutions. Nobody is going to market them for us.

All those cars people are driving today, all the machines people are using, all the phones people are buying, every one of them was once advertised. Even the dominant platforms still advertise. So how much are we investing in advertising? How much training do we have in technical marketing? This is where we must wake up. Market our solutions locally and internationally.

And government must support this. In fact I will say it more strongly, and this is the sentence I would like you to take home.

Government must be the chief marketer of local solutions and local talent, and not their competitor.

9. Six Asks: What This Conference Should Leave Behind

Let me not leave you with sentiment alone. Engineers deal in deliverables. Here are six specific things I am asking this Branch and this Society to pursue.

- One. A National Engineering Sandbox Framework. Let NSE draft and formally table a framework for supervised live testing of locally built engineering systems, modelled directly on the Central Bank's Regulatory Sandbox Framework of 2021, and addressed to NERC, NCC, ONSA, SON and COREN. If it exists for money, it can exist for machines.
- Two. Enforce Executive Order 5 of 2018. Most engineers in this room do not know that it exists. It directs every Ministry, Department and Agency to engage indigenous professionals in the planning, design and execution of national security projects, and states that a foreign professional may only be considered where it is certified that the expertise is not available in Nigeria. It has been law for eight years. Let this Branch begin by auditing federal and state engineering contracts in Oyo State against it, and publishing what it finds.
- Three. A live register of Nigerian engineering capability. NOTAP is already mandated to maintain an experts' database. Let NSE make it real, searchable

and credible, so that no procuring officer can honestly claim the capability was not available.

- Four. Ring-fence maintenance. Advocate that a defined percentage of every public infrastructure project cost, in the order of five per cent, be committed to a ten-year monitoring and maintenance obligation, with the sensing and instrumentation built in from day one rather than retrofitted after the first failure.
- Five. Open the laboratories, and insure them. A shared instrumentation scheme across the University of Ibadan, LAUTECH, the Polytechnic Ibadan and independent facilities such as RAIN, with a common insurance and access pool, so that a student can put their hands on a robot arm without a single institution bearing the entire risk of breakage.
- Six. Publish at home as well as abroad. Let every Nigerian-funded engineering paper be mirrored in an open local repository. Publish abroad for your career, by all means. But deposit at home for your country.

10. Conclusion

Distinguished ladies and gentlemen, let me bring this to a close.

The scale at which each of our stakeholders is currently producing ideas and innovations, the researchers, the students, the business owners, is too low to transform this country. That is the honest assessment. But the reason is not capability. The reason is that we have built no bridge between the laboratory and the live system. We have no sandbox.

Nigeria cannot secure what it cannot see, and cannot scale what it is never allowed to test.

We have the people. We have the market. We have the urgency. What we have been missing is a place to fail safely, a law that is actually enforced, and the confidence to buy from ourselves.

So let us look at our problems locally. Let us believe we can solve them. Let us not wait for foreign validation. Let us build the sandbox, enforce the law we already have, equip our laboratories, market what we build, and hand something over to the young people who are watching us this morning.

Thank you very much for the opportunity. I look forward to the changes that are coming in Nigeria, and to a better and more secure future. Well done on the great work we are all doing. May God give us reason to smile, for there is light at the end of the tunnel.

Thank you, and God bless you.

Appendix: Data and Sources Referenced

The following supporting facts are cited in this address. They are provided so that figures quoted from the podium can be verified and defended in the plenary session.

- Grid instability. Nigeria has recorded well over 200 partial and total grid collapses since 2010, including approximately twelve in 2024. Installed capacity is reported at roughly 13,000 to 15,500 MW, while delivered generation typically ranges between 4,000 and 5,000 MW against an estimated national requirement of about 30,000 MW. (Guardian Nigeria; BusinessDay; TCN/NISO reporting.)
- Telecoms as critical infrastructure. The Designation and Protection of Critical National Information Infrastructure Order, 2024 was gazetted on 25 June 2024, designating telecommunications assets as CNII and providing for penalties of up to ten years' imprisonment for wilful damage. Despite this, the NCC reported over 5,000 fibre cuts arising from road construction and excavation in the first half of 2026. (NCC; Office of the National Security Adviser; TechCabal; Technext.)
- Local content precedent. Nigerian content in the oil and gas industry rose from below 5% before the NOGICD Act 2010 to over 61% by 2026, with approximately US\$8 billion per annum now retained in-country. TotalEnergies reported 95% Nigerian content on a project in February 2026. (NCDMB, Nigeria Oil and Gas Energy Week 2026; AllAfrica.)
- Executive Order 5 of 2018. Formally the Presidential Executive Order for Planning and Execution of Projects, Promotion of Nigerian Content in Contracts and Science, Engineering and Technology, signed February 2018. It directs MDAs to engage indigenous professionals in national security projects and permits consideration of a foreign professional only where certified that the expertise is unavailable in Nigeria. It also mandates NOTAP to maintain an experts' database. (State House, Abuja; NIPC.)
- Regulatory sandboxes. The UK Financial Conduct Authority established the first regulatory sandbox in 2016; more than sixty countries have since adopted the model. The Central Bank of Nigeria issued its Framework for Regulatory Sandbox Operations in January 2021. Applications for Cohort 2 of the CBN Regulatory Sandbox opened on 12 August 2026 and close on 31

August 2026. The SEC operates a parallel Accelerated Regulatory Incubation Programme. (CBN; SEC; Paystack; TechCabal.)

- Technology Readiness Levels. A nine-point scale originated by NASA and adopted by the US Department of Defense, the European Commission and ISO 16290, used to classify technology maturity from basic principles (TRL 1) to a proven system in an operational environment (TRL 9). The gap at TRL 4 to 7 is commonly described in innovation policy as the valley of death.
- Applied research institutes. Germany's Fraunhofer-Gesellschaft operates on a contract research model in which a minority of income is base public funding and the majority is earned from industry and competitive projects. Taiwan's Industrial Technology Research Institute (ITRI) spun out both TSMC and UMC. The United States operates DARPA on a programme-manager-led, fixed-term model.
- Digital twins. Singapore's Virtual Singapore programme built a dynamic three-dimensional digital model of the city state. The United Kingdom's National Digital Twin Programme publishes governance principles for connected digital twins of infrastructure.
- Shadow deployment and parallel running. Standard practice in which a candidate system processes live production data in parallel with the incumbent system, with its outputs recorded and compared but never acted upon. Widely used in core banking migration, safety-critical software qualification and machine learning system rollout.